Hybrid Option Deliverables, Embedded Warrants, and Conditional Convexity in Modern Equity Markets

Abstract

This paper develops a structural research framework for analyzing how non-standard option deliverables, especially warrant-embedded deliverables, may alter conventional option exercise incentives, dealer hedge adjustments, securities lending inventory conditions, and broader reflexive market dynamics. The framework integrates five linked modules: a Structural Amplification Index (SAI), a Monte Carlo stress simulation, a gamma exposure dataset design, a borrow inventory regression specification, and an OCC option-adjustment event study architecture.

The principal claim is limited and mechanical rather than deterministic. Embedded deliverables can change the effective economics of exercise and therefore can alter hedge timing and synthetic supply dynamics. However, these effects remain conditional on empirical market inputs such as option open interest concentration, dealer positioning, borrow availability, conversion incentives, and the exact contractual structure of deliverables. Where real-world verification is required, this paper explicitly notes the missing data and unresolved inputs necessary for full empirical completion.

1 Introduction

Standard listed equity options are generally modeled as delivering one hundred shares of the underlying security per contract. This assumption is deeply embedded in conventional option pricing, exercise logic, hedge estimation, and microstructure interpretation. In most retail and professional commentary, these assumptions are treated as stable background conditions.

That simplification breaks down when the deliverable is adjusted. Corporate actions, reorganizations, distributions, special dividends, spin-offs, or warrant attachments may produce non-standard deliverables that contain additional assets beyond ordinary shares. Once that occurs, the economic object delivered at exercise is no longer equivalent to one hundred shares alone.

This matters because early exercise logic, dealer hedge management, and synthetic supply interpretation depend on what the contract actually delivers. If a contract delivers both

shares and warrants, then the option references a composite economic package. As a result, exercise decisions may diverge from the standard "never exercise slightly out-of-the-money early" intuition that is valid under ordinary assumptions.

This paper formalizes that intuition into a more rigorous framework. It does not claim automatic instability, inevitable short squeezes, or guaranteed cascades. Rather, it argues that embedded deliverables can generate conditional convexity that becomes more important when combined with concentrated gamma exposure, borrow scarcity, convertible hedge adjustment, and fragmented liquidity conditions.

2 Effective Deliverable Value

Let the adjusted option deliverable consist of:

- 100 shares of the common stock, and
- N_w warrants.

Define:

$$S = \text{spot price of the common stock,} \tag{1}$$

$$W(S) = \text{market value of one warrant as a function of } S, \tag{2}$$

$$N_w = \text{number of warrants delivered per contract.} \tag{3}$$

Then the gross economic value of the adjusted deliverable is:

$$V_{\text{deliverable}} = 100S + N_w W(S). \tag{4}$$

To express this on a per-share-equivalent basis, define the embedded uplift:

$$\Delta_{\text{embedded}} = \frac{N_w}{100} W(S). \tag{5}$$

The option can therefore be interpreted as referencing an effective synthetic underlying:

$$S^* = S + \Delta_{\text{embedded}}. \tag{6}$$

This formulation is useful because it makes clear that the contract is no longer a pure claim on the stock price alone. Instead, it is a claim on stock plus embedded contingent value.

3 Modified Early Exercise Condition

Under a standard call option framework without dividends and without embedded side-assets, early exercise is typically suboptimal when the option retains time value. That result depends on the exercise decision comparing intrinsic value to the value of maintaining optionality.

With warrant-embedded deliverables, the exercise decision changes. A useful accounting identity for the early exercise advantage is:

$$EEA = 100(S - K) + N_w W(S) - P_{\text{remaining}} - C, \tag{7}$$

where:

$$K = \text{strike price}, \tag{8}$$

$$P_{\text{remaining}} = \text{remaining option premium or foregone optionality value}, \tag{9}$$

$$C = \text{transaction costs, financing, and operational frictions.} \tag{10}$$

Exercise is economically rational when:

$$EEA > 0. \tag{11}$$

The key implication is that exercise may be rational even when $S < K$, provided the warrant value is sufficiently large relative to the out-of-the-money amount plus remaining premium and frictions. This does not mean such exercise always occurs. It means the standard early exercise heuristic is no longer mechanically reliable once the deliverable includes meaningful additional assets.

4 Dealer Hedging Implications

Dealer hedge behavior is affected by whether exposure remains in derivative form or is converted into physical delivery obligations. Let:

$$H_t = \text{dealer hedge inventory at time } t. \tag{12}$$

If exercise removes a block of option open interest from the chain, then a stylized update equation is:

$$H_{t+1} = H_t - \Delta_{\text{exercise}}, \tag{13}$$

where Δ_{exercise} captures the hedge adjustment associated with exercised contracts.

At the market level, exercise can:

- reduce open interest,
- alter dealer delta exposure,
- change gamma concentration at nearby strikes,
- shift the balance between derivative exposure and physical delivery obligations.

If enough contracts are exercised for embedded-asset extraction rather than retained as pure options exposure, the options market can lose some of the synthetic layering that previously mediated hedge flows.

5 Convertible Arbitrage Interaction

A standard stylized convertible arbitrage position can be written as:

$$\text{Long convertible bond} + \text{short stock}. \tag{14}$$

Let the hedge ratio be:

$$H_{\text{convert}} = Q \cdot \Delta_{\text{convert}}, \tag{15}$$

where:

$$Q = \text{share-equivalent conversion quantity}, \tag{16}$$

$$\Delta_{\text{convert}} = \text{convertible delta}. \tag{17}$$

As the underlying stock price rises, Δ_{convert} typically increases, which can reduce the need for a large short stock hedge or force hedge rebalancing. Under certain price regimes, that can create buy-to-cover flow.

This paper does not assume a specific active convertible arbitrage overhang without documentary confirmation. Rather, it includes this module because embedded deliverable incentives may interact with existing convertible hedge structures when both are present in the same security.

6 Gamma Reflexivity

Dealer hedge demand can become nonlinear when gamma exposure is large and concentrated. A stylized local approximation is:

$$\Delta\text{Hedge} \approx \Gamma \cdot \Delta S, \tag{18}$$

where:

$$\Gamma = \text{aggregate gamma exposure in the relevant price region,} \tag{19}$$

$$\Delta S = \text{change in underlying price.} \tag{20}$$

When gamma is concentrated near major strikes and expirations, even modest price moves can force repeated hedge adjustments. This can amplify realized intraday movement. It does not determine direction by itself, but it increases sensitivity to flow.

In a market with embedded deliverable distortions, the gamma profile may change not only because price changes, but because exercise behavior changes which contracts remain open.

7 Borrow Inventory Dynamics

Securities lending is best understood as an inventory system rather than a static stock. Let:

$$L_t = \text{lendable share inventory at time } t. \tag{21}$$

A stylized inventory transition is:

$$L_{t+1} = L_t - R_t - E_t + \Lambda_t, \tag{22}$$

where:

$$R_t = \text{share recalls or internal inventory withdrawals,} \tag{23}$$

$$E_t = \text{exercise-related physical share removals,} \tag{24}$$

$$\Lambda_t = \text{new lendable supply entering the system.} \tag{25}$$

If inventory exits exceed replenishment, borrow tightness rises, utilization increases, and the cost to borrow may move higher. That does not force immediate covering in all cases, but it changes the economics of maintaining short or short-equivalent positions.

8 Structural Amplification Index

To operationalize the framework, define a Structural Amplification Index (SAI) as a weighted composite of the main reflexive stress variables:

$$SAI_t = w_1\Gamma_t^{\text{norm}} + w_2 B_t^{\text{norm}} + w_3 OI_t^{\text{norm}} + w_4 M_t^{\text{norm}} + w_5 X_t^{\text{norm}}, \tag{26}$$

subject to:

$$\sum_{i=1}^{5} w_i = 1, \qquad w_i \geq 0. \tag{27}$$

The normalized components are:

$$\Gamma_t^{\text{norm}} = \text{normalized gamma concentration or net gamma measure,} \tag{28}$$

$$B_t^{\text{norm}} = \text{normalized borrow stress or utilization measure,} \tag{29}$$

$$OI_t^{\text{norm}} = \text{normalized open interest concentration measure,} \tag{30}$$

$$M_t^{\text{norm}} = \text{normalized short-horizon price momentum or realized volatility proxy,} \tag{31}$$

$$X_t^{\text{norm}} = \text{optional event or exercise-pressure term.} \tag{32}$$

A practical implementation could set:

$$X_t^{\text{norm}} = \begin{cases} 1 & \text{if a verified adjustment, warrant extraction window, or delivery event is active,} \\ 0 & \text{otherwise.} \end{cases} \tag{33}$$

8.1 Interpretation

Higher SAI values indicate a greater likelihood that local market flows may become structurally amplified. The index is not a forecast of price direction. It is a regime indicator designed to identify when reflexive microstructure conditions are more fragile.

8.2 What Is Missing for Full Completion

The SAI framework is structurally complete, but full empirical construction still requires:

- historical option chain data by strike and expiry,
- a real borrow utilization and cost-to-borrow dataset,
- a documented event series for deliverable adjustments,
- a calibration method for the weights w_i.

9 Monte Carlo Stress Simulation

To model how reflexive stress may develop under different market states, consider a Monte Carlo system with both stochastic price evolution and endogenous feedback from hedging and borrow conditions.

A baseline continuous-time price process can be written:

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t + \phi_t \, dt, \tag{34}$$

where:

$$\mu = \text{drift}, \tag{35}$$
$$\sigma = \text{volatility}, \tag{36}$$
$$dW_t = \text{Brownian shock}, \tag{37}$$
$$\phi_t = \text{endogenous flow term from hedging and inventory feedback.} \tag{38}$$

A discrete-time implementation may define:

$$S_{t+1} = S_t \exp\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t}\,\varepsilon_t + \kappa \cdot SAI_t\right], \tag{39}$$

where $\varepsilon_t \sim N(0,1)$ and κ scales the effect of structural stress.

The lending inventory may be simulated simultaneously:

$$L_{t+1} = L_t - R_t - E_t + \Lambda_t, \tag{40}$$

with event-driven increases in E_t during hypothetical exercise waves and reductions in Λ_t under tighter inventory conditions.

9.1 Simulation Outputs

A complete simulation should produce:

- distributions of future price paths,
- pathwise gamma shock intensity,
- borrow inventory decay trajectories,
- threshold frequencies for high-SAI regimes,
- event-conditioned stress probabilities.

9.2 What Is Missing for Full Completion

The simulation design is complete, but the following empirical inputs are still missing:

- estimated or observed gamma concentration parameters,

- historical volatility inputs and event-specific regime adjustments,
- actual borrow inventory series,
- assumptions about exercise-trigger probabilities,
- conversion or adjustment event dates from primary sources.

10 Gamma Exposure Dataset

A gamma exposure dataset is necessary to estimate whether dealers are likely to buy strength or sell strength around key strikes.

A minimum dataset schema should include:

Field	Description
date	observation date
ticker	underlying security
spot_price	closing or intraday reference price
strike	option strike
expiration	expiration date
option_type	call or put
open_interest	open contracts outstanding
implied_volatility	implied volatility input
delta	option delta
gamma	option gamma
contract_multiplier	usually 100, unless adjusted
adjusted_deliverable_flag	indicator for non-standard deliverable
deliverable_notes	textual description of deliverable terms

A stylized gamma exposure measure at strike k is:

$$GEX_k = OI_k \cdot Mult_k \cdot \Gamma_k \cdot S^2, \tag{41}$$

where:

$$OI_k = \text{open interest at strike } k, \tag{42}$$

$$Mult_k = \text{contract multiplier}, \tag{43}$$

$$\Gamma_k = \text{option gamma at strike } k. \tag{44}$$

Aggregate gamma exposure is then:

$$GEX_{\text{total}} = \sum_k GEX_k. \tag{45}$$

10.1 What Is Missing for Full Completion

The dataset design is ready, but the empirical module remains incomplete without:

- historical option chain data,
- a consistent gamma calculation methodology,
- documented treatment of adjusted contracts,
- a dealer sign convention for whether dealers are assumed net short or net long options.

11 Borrow Inventory Regression

To test whether inventory stress explains changes in borrow cost, define a baseline panel or time-series regression:

$$BorrowRate_t = \beta_0 + \beta_1 Utilization_t + \beta_2 ShortInterest_t + \beta_3 Momentum_t + \beta_4 Volatility_t + \beta_5 EventDummy_t + \tag{46}$$

The variables are:

$$BorrowRate_t = \text{observed cost-to-borrow or fee}, \tag{47}$$
$$Utilization_t = \text{shares on loan divided by lendable inventory}, \tag{48}$$
$$ShortInterest_t = \text{short interest measure}, \tag{49}$$
$$Momentum_t = \text{recent return or price acceleration measure}, \tag{50}$$
$$Volatility_t = \text{realized or implied volatility measure}, \tag{51}$$
$$EventDummy_t = \text{indicator for adjustment or delivery-related events}. \tag{52}$$

11.1 Interpretation

A positive and significant estimate for β_1 would support the claim that tighter borrow inventory conditions are associated with higher borrow cost. A positive β_5 would support the hypothesis that special events, including deliverable adjustments or exercise-driven inventory drain, coincide with elevated borrow pressure.

11.2 What Is Missing for Full Completion

This model cannot be run yet because the following are still missing:

- a time series of cost-to-borrow data,
- lendable inventory or utilization data,
- an event calendar tied to actual contract adjustments,
- a decision on estimation design: OLS, fixed effects, distributed lags, or regime-switching specification.

12 OCC Event Study

An event study is needed to test whether option-adjustment announcements or deliverable redefinitions are associated with measurable changes in volatility, open interest structure, gamma exposure, or borrow conditions.

Let event date $t = 0$ denote the verified publication of an OCC adjustment memo or equivalent official deliverable notice. A baseline event window may be:

$$t \in [-5, +10] \tag{53}$$

trading days around the event.

Potential dependent variables include:

- abnormal stock returns,
- changes in implied volatility,
- changes in open interest concentration,
- changes in aggregate gamma exposure,
- changes in borrow availability or fee.

A simple event-regression form is:

$$Y_t = \alpha + \beta \cdot EventWindow_t + \gamma Z_t + \varepsilon_t, \tag{54}$$

where Y_t is the chosen market outcome variable and Z_t contains control variables such as market return, realized volatility, or broad meme-stock sector movement.

12.1 What Is Missing for Full Completion

This module remains incomplete until the following are obtained:

- the exact OCC memo or official notice,
- a verified event date series,
- pre- and post-event options chain data,
- borrow data aligned to the event window,
- a benchmark model for abnormal return or abnormal volatility estimation.

13 Conditional Convexity Regime

The framework suggests that market instability is conditional rather than automatic. The strongest amplification is expected only when multiple layers align simultaneously:

1. high open interest concentration near relevant strikes,
2. elevated gamma sensitivity,
3. rising or accelerating underlying price,
4. meaningful embedded warrant value relative to remaining option premium,
5. declining lendable inventory,
6. possibly, convertible hedge unwind or event-driven delivery stress.

Without this overlap, the market may remain well behaved even when one individual metric looks extreme. The main contribution of the framework is therefore diagnostic. It helps distinguish isolated stress from multi-variable reflexive stress.

14 Data Requirements and Completion Checklist

14.1 Structurally Complete in This Draft

The following components are conceptually complete:

- effective deliverable valuation framework,
- modified early exercise condition,
- dealer hedge adjustment logic,

- convertible interaction module,
- gamma reflexivity framework,
- borrow inventory accounting identity,
- Structural Amplification Index definition,
- Monte Carlo architecture,
- gamma exposure dataset schema,
- borrow regression specification,
- OCC event study design.

14.2 Still Missing and Required for Full Empirical Completion

The following items remain unresolved and must be filled from primary or market datasets:

- exact OCC deliverable memo confirming the adjusted contract terms,
- exact warrants-per-contract ratio,
- warrant strike and expiration details,
- historical option chain data by strike and expiry,
- methodology for dealer gamma sign assumptions,
- securities lending inventory and utilization data,
- cost-to-borrow time series,
- event calendar for adjustments and contract redefinitions,
- any verified convertible prospectus terms if a convert module is to be estimated directly.

15 Limitations

This paper is a structural framework, not a completed empirical finding. Several market layers discussed here are inherently difficult to observe with perfect precision, especially dealer inventory, lending internalization, and contract-specific exercise behavior.

The framework therefore has three important limitations. First, some key variables are only partially observable in public data. Second, the exact magnitude of any reflexive effect depends on contract terms that must be verified from official documentation. Third, even

a correctly specified structural model does not imply inevitability; conditional convexity is not the same thing as guaranteed cascade dynamics.

16 Conclusion

Non-standard option deliverables can alter the economics of exercise in ways that standard equity option heuristics do not capture. When an option delivers both shares and warrants, the relevant exercise decision becomes a composite-asset decision rather than a simple share-acquisition decision.

That change can affect dealer hedge demand, open interest decay, synthetic supply interpretation, and securities lending stress. When combined with concentrated gamma exposure, rising price momentum, and tighter lendable inventory, these features can produce conditional convexity and reflexive amplification.

At present, the framework is substantially complete on theory and model architecture but incomplete on empirical verification. The next step is not further narrative expansion. It is dataset completion: OCC documentation, historical option chain reconstruction, borrow inventory collection, and event-window estimation. Once those inputs are assembled, the framework can be converted from a defensible structural paper into a testable empirical microstructure study.